

Agniveer Vayu Science (Group X) - 14 Oct 2023 Shift 1 - Memory Based Paper

Physics

Question 1

The angle of incidence at which reflected light is polarized for reflection from air to glass (refraction index μ) is

Options:

- A. $\sin^{-1}(\mu)$
- B. $\sin^{-1}(1/\mu)$
- C. $\tan^{-1}(\mu)$
- D. $\tan^{-1}(1/\mu)$

Answer: C

Solution:

CONCEPT:

Brewster angle –

- The angle of incidence at which a beam of unpolarised light falling on a transparent surface is reflected as a beam of completely plane polarised light is called **polarising or Brewster angle**. It is denoted by i_p .
- The British Physicist David Brewster found the relationship between Brewster angle (i_p) and refractive index (μ) –

$$\mu = \tan i_p$$

- This relation is known as **Brewster law**.

Explanation:

Given - refraction index = μ

We know, $\mu = \tan i_p$

Where i_p = Polarization or Brewster's angle

$$\therefore i_p = \tan^{-1}(\mu)$$

Question 2

Potential inside a hollow sphere is:

Options:

- A. constant
- B. proportional to distance from centre
- C. inversely proportional to the distance
- D. inversely proportional to square of distance

Answer: A

Solution:

CONCEPT:

Electrical potential:

- The **electric potential** at any point in the **electric field** is defined as the **amount of work done** in moving a **unit positive test charge** from **infinity to that point** without **acceleration**.

$$\Rightarrow V = \frac{kq}{r}$$

Electrical potential difference:

- In an electrical field, the potential difference between two points (V) is defined as the amount of work done (W) by an external agent in moving a unit charge from one point to another.
- It is given as:

$$\Rightarrow \Delta V = \frac{W}{Q}$$

Where W = Work done, Q = Charge, ΔV = Electrical potential difference, V = electric potential at a point, $k = 9 \times 10^9 \text{ N-m}^2/\text{C}^2$, q = charge, and r = distance of a point from the center of the charge

CALCULATION:

- Since in a **hollow sphere**, the **charge** is only present at **the surface**, so the **electric field intensity inside the hollow sphere is zero**.

- So it is not required to do work done on a unit positive test charge from one point to another inside the hollow sphere.
 - Hence **potential inside a hollow sphere is constant** ($\Rightarrow V = \frac{kq}{r}$).
 - Hence, option 1 is correct.
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Question 3

Moment of inertia depends on:

Options:

- A. Mass
- B. Distribution of mass
- C. Angular velocity
- D. Position of rotational axis and mass distribution

Answer: D

Solution:

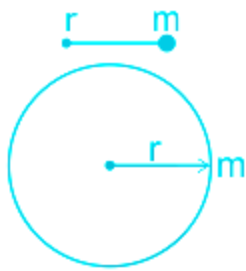
Concept:

- **Moment of Inertia:** A quantity expressing a body's tendency to resist angular acceleration is called the moment of Inertia.
- For point mass **Moment of inertia** is simply the mass times the square of the perpendicular distance to the axis of rotation.

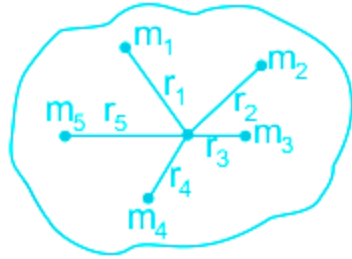
$$\Rightarrow I = m \times r^2$$

where I is the Moment of Inertia, m is point mass, r is the perpendicular distance from the axis of rotation.

- For a **rigid body system**, the **moment of inertia** is the sum of the moments of inertia of all its particles taken about the same axis.



$$I = mr^2$$



$$I = \sum m_i r_i^2 = m_1 r_1^2 + m_2 r_2^2 + m_3 r_3^2 + m_4 r_4^2 + m_5 r_5^2$$

$$\Rightarrow I = \sum m_i r_i^2$$

where I is the Moment of Inertia, m is point mass, and r is the perpendicular distance from the axis of rotation.

Explanation:

- If the mass is situated close to the axis Moment of Inertia will be small because the distance of mass particles from the axis of rotation will be small.
- If the mass is situated at a large distance from the axis Moment of Inertia will be large because the distance of mass particles from the axis of rotation will be large.
- So the moment of Inertia depends on the position of the axis of rotation as well as the mass distribution.
- The moment of inertia is a physical property of configuration mass distribution and an axis of rotation.
- But there is no role of the angular velocity to find the moment of inertia. (see formula also)

Question 4

In a common emitter transistor the current gain β equals to _____, where I_b , I_c , I_e have their usual meanings.

Options:

A. I_b/I_c

B. I_e/I_c

C. I_c/I_b

D. I_c/I_e

Answer: C

Solution:

CONCEPT:

- **Transistor:** A semiconductor device used to amplify or switch electronic signals and electrical power is called a transistor.
 - It is made of semiconductor material with at least three terminals for connection to an external circuit.
- **Alpha gain (α):** In the common-base configuration, the current gain that is defined as the ratio of change in collector current to change in emitter current is known as alpha gain.

$$\alpha = \frac{I_c}{I_e}$$

- **Beta gain (β):** It is the current gain in common Emitter configuration. The current gain that is defined as the change in collector current to base current is known as beta gain.

$$\beta = \frac{I_c}{I_b}$$

EXPLANATION:

- In a common-emitter transistor, the current gain β equals to

$$\Rightarrow \beta = \frac{I_c}{I_b}$$

NOTE:

- The relation between alpha gain and beta gain is given by:

$$\alpha = \frac{\beta}{1+\beta}$$

where α is the alpha gain and β is the beta gain

Question 5

SI unit of permittivity is _____

Options:

- A. $C^2 m^2 N^{-1}$
- B. $C^{-1} m^2 N^{-2}$
- C. $C^2 m^2 N^2$
- D. $C^2 m^{-2} N^{-1}$

Answer: D

Solution:

CONCEPT:

- **Coulomb's law in Electrostatics:** It states that **force of interaction** between two stationary point charges is **directly proportional** to the **product of the charges**, and **inversely proportional** to the **square of the distance** between them and acts along the straight line joining the two charges.



$$F \propto q_1 \times q_2$$

$$F \propto \frac{1}{r^2}$$

$$F = K \frac{q_1 \times q_2}{r^2}$$

Where $K = \frac{1}{4\pi\epsilon_0}$ constant called **electrostatic force constant**.

- The **value of K** depends on the **nature of the medium** between the two charges and the system of units chosen

EXPLANATION:

- From above it is clear that the **constant in Coulomb's Law equation** is $\frac{1}{4\pi\epsilon_0}$, where ϵ_0 is the **permittivity of free space**. Therefore Coulomb's law in Electrostatics can be written as

$$\Rightarrow F = \frac{1}{4\pi\epsilon_0} \frac{q_1 \times q_2}{r^2}$$

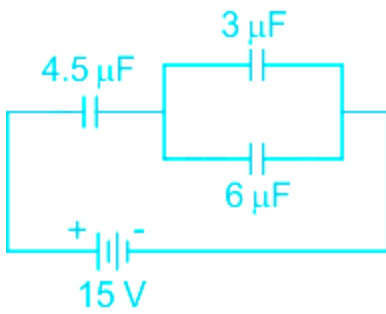
The above equation can be written as,

$$\Rightarrow \epsilon_0 = \frac{q_1 q_2}{F r^2} = \frac{C^2}{Nm^2}$$

- The **SI unit of permittivity** is $C^2 N^{-1} m^{-2}$. Therefore option 4 is correct.

Question 6

The potential difference across the $4.5\mu F$ capacitor in the given circuit is



Options:

A. 10 V

B. 8 V

C. 15 V

D. 7 V

Answer: A

Solution:

The correct answer is option 1) i.e. 10 V

CONCEPT:

- **Capacitor:** A capacitor is an electrical component with two terminals used to store charge in the form of an electrostatic field.
 - It consists of two parallel plates each possessing equal and opposite charges, separated by a dielectric constant.
 - **Capacitance is the ability of a capacitor to store charge in it.** The capacitance C is related to the charge Q and voltage V across them as:

$$\Rightarrow C = \frac{Q}{V}$$

- **Equivalent capacitance of capacitors-**

- **Connected in series:** When n capacitors $C_1, C_2, C_3, \dots, C_n$ are connected in series, the net capacitance (C_s) is given by:

$$\Rightarrow \frac{1}{C_s} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + \dots + \frac{1}{C_n}$$

- **Connected in parallel:** When n capacitors $C_1, C_2, C_3, \dots, C_n$ are connected in parallel, the net capacitance (C_p) is given by:

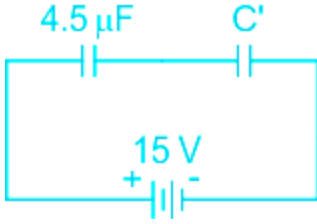
$$\Rightarrow C_p = C_1 + C_2 + C_3 + \dots + C_n$$

CALCULATION:

Given that:

Voltage, $V = 15 \text{ V}$

Capacitors $3\mu\text{F}$ and $6\mu\text{F}$ are in parallel. Hence, their equivalent capacitance $C' = 6 + 3 = 9\mu\text{F}$



C' and $4.5\mu\text{F}$ are in series. Hence, the **equivalent capacitance of the circuit** is $\frac{1}{C''} = \frac{1}{4.5} + \frac{1}{9} = \frac{9+4.5}{9 \times 4.5} = \frac{1}{3}$

$$\Rightarrow C'' = 3\mu\text{F}$$

Total charge in the circuit, $Q = CV = 3 \times 15 = 45 \mu\text{C}$

To find the potential difference across $4.5\mu\text{C}$:

$4.5\mu\text{C}$ and C' are in series. **The charge across capacitances in series is constant.**

Using, $C = \frac{Q}{V}$

$$\Rightarrow V = \frac{Q}{C} = \frac{45}{4.5} = 10 \text{ V}$$

Question 7

Consider an infinitely long straight conductor carrying current I . The magnetic field B due to this current-carrying conductor at a distance d from it is

Options:

A. $\frac{\mu_0 I}{\pi d}$

B. $\frac{\mu_0 I}{d}$

C. $\frac{\mu_0 I}{2\pi d}$

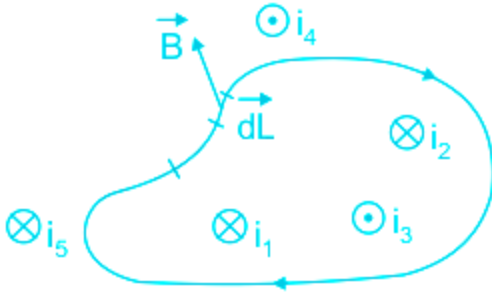
D. $\frac{2\pi\mu_0 I}{d}$

Answer: C

Solution:

CONCEPT:

Ampere's law



- The circulation $\oint \mathbf{B} \cdot d\mathbf{l}$ of the resultant magnetic field along a closed, plane curve is equal to μ_0 times the total current crossing the area bounded by the closed curve provided the electric field inside the loop remains constant.

$$\Rightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$

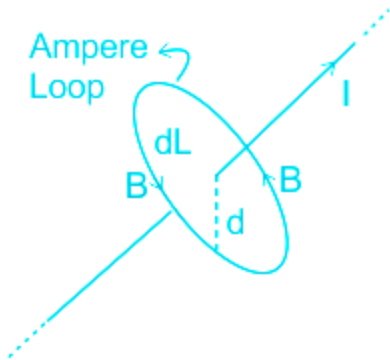
where B is the magnetic field at the length element $d\mathbf{l}$ of the loop. I is the sum of all currents inside the loop.

- The sign convention is decided by the right-hand thumb rule. Since the loop is clockwise, curl your right-hand fingers along the loop, and the thumb will give the positive direction of current (inside the plane of the paper). Thus i_1 and i_2 are taken as positive, while i_3 is taken as negative. Here i_4 and i_5 are ignored since they are outside the loop.
- B is the net magnetic field on the loop element.

EXPLANATION:

According to Ampere's law

$$\Rightarrow \oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$$



- The magnetic field is always parallel to the element $d\mathbf{l}$ on the

$$\Rightarrow B \cdot 2\pi d = \mu_0 I$$

$$\Rightarrow B = \frac{\mu_0 I}{2\pi d}$$

- Therefore option 3 is correct.
-

Question 8

The mechanical equivalent of heat

Options:

- A. has same dimension as work
- B. has same dimension as heat
- C. has same dimension as rate of doing work
- D. dimensionless

Answer: D

Solution:

Explanation:

Mechanical equivalent of heat (J): It is defined as the amount of work needed to produce 1 calorie of heat.

or Mechanical equivalent of heat states that 4.1868 J of mechanical energy or work is required to raise the temperature of 1g of water for 1°C change in temperature.

It is just a conversion factor between two different units of energy i.e. calorie to the joule.

$$(\Sigma W)_{\text{cycle}} = J (\Sigma Q)_{\text{cycle}}$$

Where J is the mechanical equivalent of heat

since 1 calorie of heat = 4.1868 Joule of work

Or, 1 kcal of heat = 4.1868 kJ of work

Thus, 4.1868 kJ = J × 1 kcal

Hence, J = 4.1868 kJ/kcal

As it is a conversion factor for different system of units of energy, hence it is dimensionless.

Question 9

A parallel plate capacitor has a capacitance of 10μF. If the distance between two plates is doubled then the new capacitance will be-

Options:

- A. $20\mu\text{F}$
- B. $15\mu\text{F}$
- C. $10\mu\text{F}$
- D. $5\mu\text{F}$

Answer: D

Solution:

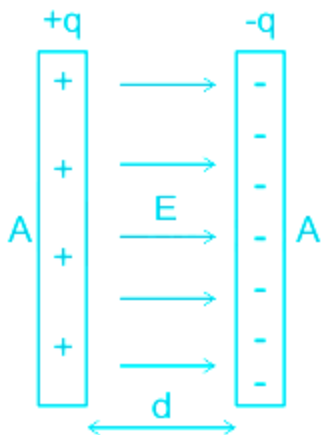
CONCEPT:

- **The capacitance of a capacitor (C):** The capacitance of a conductor is the ratio of **charge (Q)** to it by a rise in its **potential (V)**, i.e.

$$C = Q/V$$

For a Parallel Plate Capacitor:

- A **parallel plate capacitor** consists of two large plane parallel conducting plates of area A and separated by a small distance d .



- Mathematical expression for the capacitance of the parallel plate capacitor is given by:

$$\Rightarrow C = \frac{\epsilon_0 A}{d}$$

Where C = capacitance, A = area of the two plates, ϵ = dielectric constant (simplified!), d = separation between the plates.

- The unit of **capacitance** is the **farad**, (symbol **F**).

EXPLANATION:

Given - $C_1 = 10\mu\text{F}$ and $d_1 = 2d$

- The capacitance of the parallel plate capacitor is given by:

$$\Rightarrow 10 \mu F = \frac{\epsilon_o A}{d} \text{ ----- (1)}$$

- The capacitance of the parallel plate capacitor when the distance between the plate is given by:

$$\Rightarrow C_2 = \frac{\epsilon_o A}{d_1} = \frac{\epsilon_o A}{2d} \text{ ----- (2)}$$

On dividing equation 1 and 2, we get

$$\Rightarrow \frac{10 \mu F}{C_2} = \frac{\frac{\epsilon_o A}{d}}{\frac{\epsilon_o A}{2d}} = 2$$

$$\Rightarrow C_2 = 10/2 = 5 \mu F$$

Question 10

If the length of the semi-major axis of the elliptical orbit of a planet is doubled, the relation between the new time period of revolution of the planet around the Sun with the original time period will be-

Options:

A. $T_2^2 = T_1^2$

B. $T_2^2 = 2 \times T_1^2$

C. $T_2^2 = 4 \times T_1^2$

D. $T_2^2 = 8 \times T_1^2$

Answer: D

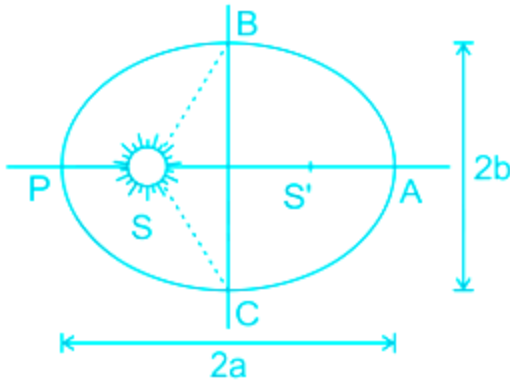
Solution:

The correct answer is **option 4)** i.e. $T_2^2 = 8 \times T_1^2$

CONCEPT:

- **Kepler's laws of planetary motion:** Kepler's laws of planetary motion can be stated as:
 - **Kepler's first law:** All the planets move around the Sun in elliptical orbits having the Sun at one of the foci. This law is also called the "**Law of orbits.**"
 - **Kepler's second law:** The radius vector drawn from the Sun to the planet sweeps out equal areas in equal intervals of time. This law is also called the "**Law of Equal Areas.**"

- **Kepler's third law:** The square of the time period of revolution of the planet around the Sun in an elliptical orbit is directly proportional to the cube of its **semi-major axis**. This law is also called the "**Law of Periods**."



According to **Kepler's third law** of planetary motion:

$$T^2 \propto a^3$$

where T = Time period of revolution of the planet, and a = length of the semi-major axis of the elliptical orbit of the planet.

CALCULATION:

The **original time period**(T_1) is related to the length of the **semi-major axis**(a_1) as:

$$T_1^2 \propto a_1^3$$

On doubling the length of the **semi-major axis**, the **new time period**(T_2) will be:

$$T_2^2 \propto (2 \times a_1)^3$$

$$\Rightarrow T_2^2 \propto 8 \times a_1^3$$

$$\Rightarrow T_2^2 = 8 \times T_1^2$$

Hence, on doubling the length of the semi-major axis, the **square of the new time period will be 8 times the square of the original time period**.

Question 11

The velocity of a wave is 700 m/s and the frequency is 28Hz. Find the wavelength of the wave.

Options:

A. 25 cm

- B. 25 m
- C. 50 cm
- D. 50 m

Answer: B

Solution:

CONCEPT:

- **Wave** is a disturbance that transfers energy from one place to another.
- The distance between one crest and the other crest or the distance between one trough to another is called a **wavelength of the wave**.
- **Wavelength** is denoted by λ .
- The number of waves that pass through a fixed point in a given time is called a **frequency** of the wave. It is denoted by f .
- The relation between wavelength and the frequency of the wave is;

$$V = f\lambda$$

Where f is frequency, λ is the wavelength and V is the speed of the wave.

EXPLANATION:

Given that; Speed = $V = 700$ m/s Frequency = $f = 28$ Hz

- So wavelength,

$$\Rightarrow \lambda = V/f = 700/28 = 25\text{m}$$

So, the correct answer is **option 2**

Question 12

The efficiency of the Carnot engine is

Options:

A. $1 - \frac{T_1}{T_2}$

B. $1 - \frac{T_2}{T_1}$

C. $1 + \frac{T_2}{T_1}$

D. $1 + \frac{T_1}{T_2}$

Answer: B

Solution:

Concept:

The efficiency of the Carnot cycle (η) –

It is defined as the ratio of net mechanical work done per cycle the gas (W) to the amount of heat energy absorbed per cycle from the source (Q_1).

$$\text{i.e. } \eta = \frac{W}{Q_1}$$

As work done by the engine per cycle is

$$W = Q_1 - Q_2$$

Where, Q_1 = amount of heat energy absorbed per cycle from the source and Q_2 = energy absorbed per cycle from the sink.

$$\therefore \eta = \frac{Q_1 - Q_2}{Q_1} = 1 - \frac{Q_2}{Q_1}$$

$$\text{As } \frac{Q_2}{Q_1} = \frac{T_2}{T_1}$$

$$\therefore \eta = 1 - \frac{T_2}{T_1}$$

Where T_1 = temperature of the source and T_2 = temperature of the sink.

Explanation-

The efficiency of the Carnot engine:

$$\eta = \frac{W}{Q_1} = \frac{Q_1 - Q_2}{Q_1} = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

Therefore the **Efficiency of the Carnot cycle** is $\eta = 1 - \frac{T_2}{T_1}$.

Mathematics

Question 13

Find the harmonic mean of 3, 6 and 10.

Options:

A. 6.33

B. 5

C. 5.6

D. 4.5

Answer: B

Solution:

Given

Observations = 3, 6, 10

Formula used

Harmonic mean = $n / \sum 1/x_i$ where $i = 1, 2, 3, 4, \dots$

n = number of observation

x = observations value

Calculation

$x_i = 3, 6, 10$

$$\Rightarrow \sum 1/x_i = 1/3 + 1/6 + 1/10 = 3/5$$

$$\Rightarrow \text{number of observation} = 3$$

$$\Rightarrow \text{Harmonic mean} = 3 / (3/5) = (3 \times 5) / 3$$

$\therefore$ Harmonic mean = 5

Important Points

The harmonic mean is defined as the reciprocal of the arithmetic mean of the observations of a set

Harmonic mean = $n / \sum 1/x_i$ where $i = 1, 2, 3, 4, \dots$

n = number of observation

x = observations value

Geometric mean = it is defined as the n -th root of the product of the n observations.

$$GM = n(\sqrt[n]{x_1 \times x_2 \times \dots \times x_n})$$

Question 14

If $3 \tan^{-1}x + \cot^{-1}x = \pi$, then x equals

Options:

A. 0

B. 1

C. -1

D. $\frac{1}{2}$

Answer: B

Solution:

Explanation:

We have, $3 \tan^{-1}x + \cot^{-1}x = \pi$

$\Rightarrow 2 \tan^{-1}x + \tan^{-1}x + \cot^{-1}x = \pi$

$\Rightarrow 2 \tan^{-1}x + \pi/2 = \pi$

$\Rightarrow 2 \tan^{-1}x = \pi/2$

$\Rightarrow \tan^{-1}x = \pi/4$

$\Rightarrow x = \tan(\pi/4) = 1$

$\therefore x = 1$

Question 15

Find the equation of directrix of the parabola $y^2 + 8y - 12x + 4 = 0$?

Options:

A. $x - 4 = 0$

B. $x + 4 = 0$

C. $x + 2 = 0$

D. $x - 2 = 0$

Answer: B

Solution:

Concept:

General equation of Parabola $(y - k)^2 = 4a(x - h)$ and directrix of parabola $y^2 = 4ax$ is $x = -a$.

Calculation:

The equation of parabola is $y^2 + 8y - 12x + 4 = 0$

$$\Rightarrow y^2 + 8y + 16 - 16 - 12x + 4 = 0$$

$$\Rightarrow (y + 4)^2 - 16 - 12x + 4 = 0$$

$$\Rightarrow (y + 4)^2 = 12x + 16 - 4$$

$$\Rightarrow (y + 4)^2 = 12x + 12$$

$$\Rightarrow (y + 4)^2 = 12(x + 1)$$

Let $y + 4 = Y$, $x + 1 = X$ (i)

$$\therefore Y^2 = 12X$$

Comparing it with $Y^2 = 4aX$ we get

$$\therefore a = 3 \text{(ii)}$$

As we know that directrix of parabola $Y^2 = 4aX$ is given by $X = -a$.

From equation i and ii.

$$x + 1 = -3$$

$$\Rightarrow x + 4 = 0$$

So, equation of directrix is $x + 4 = 0$

Hence, **option 2** is correct.

Question 16

Find the value of k for which the function

$$f(x) = \begin{cases} kx + 5, & \text{when } x \leq 2 \\ x - 1, & \text{when } x > 2 \end{cases} \text{ is continuous at } x = 2?$$

Options:

A. -2

B. 2

C. 1

D. -1

Answer: A

Solution:

Concept:

Continuity:

For a function say f, $\lim_{x \rightarrow a} f(x)$ exists

$$\Rightarrow \lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = l = \lim_{x \rightarrow a} f(x), \text{ where } l \text{ is a finite value.}$$

Any function say f is said to be **continuous** at point say a if and only if $\lim_{x \rightarrow a} f(x) = l = f(a)$, where l is a finite value.

Calculation:

$$\text{Given: The function } f(x) = \begin{cases} kx + 5, & \text{when } x \leq 2 \\ x - 1, & \text{when } x > 2 \end{cases} \text{ is continuous at } x = 2.$$

$\therefore f(x)$ is continuous at $x = 2$.

As we know that, if a function say f is said to be **continuous** at point say a if and only if $\lim_{x \rightarrow a} f(x) = l = f(a)$, where l is a finite value.

$$\Rightarrow \lim_{x \rightarrow 2} f(x) = f(2)$$

Left hand Limit: **LHL**

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} f(2 - h) = \lim_{h \rightarrow 0} \{k \cdot (2 - h) + 5\} = 2k + 5$$

Right hand Limit: **RHL**

$$\lim_{x \rightarrow 2^+} f(x) = \lim_{x \rightarrow 2^+} f(2+h) = \lim_{h \rightarrow 0} \{(2+h) - 1\} = 1$$

$\therefore \lim_{x \rightarrow 2} f(x)$ exists

$$\Rightarrow \lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^+} f(x)$$

$$\text{So, } 2k + 5 = 1$$

$$\Rightarrow k = -2.$$

Hence, $f(x)$ is continuous at $x = 2$ when $k = -2$.

Question 17

Thirty two number card bearing numbers 1 to 32 are put in a box and then a number card is drawn from it at random.

What is the probability that the number card bears a prime number?

Options:

A. $\frac{13}{32}$

B. $\frac{3}{8}$

C. $\frac{11}{32}$

D. $\frac{5}{16}$

Answer: C

Solution:

Given:

Total number of cards = 32

Formula used:

Probability of an event = Total number of favorable outcomes/Total number of outcomes

Calculation:

Total number of cards = 32

∴ Total number of outcomes = 32

Prime numbers between 1 to 32 = 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31

∴ Total number of favorable outcomes = 11

The probability that the number card bears a prime number = $11/32$

∴ The probability that the number card bears a prime number is $11/32$

Question 18

$$\int x^3 e^{x^2} dx =$$

Options:

A. $\frac{1}{2}(x^2 + 1)e^{x^2} + c$

B. $(x^2 + 1)e^{x^2} + c$

C. $\frac{1}{2}(x^2 - 1)e^{x^2} + c$

D. $(x^2 - 1)e^{x^2} + c$

Answer: C

Solution:

Concept:

Integration by parts-

$$\int u \cdot v \, dx = u \cdot \int v \, dx - \int (u' \cdot \int v \, dx) \, dx$$

Calculation:

Given, $I = \int x^3 e^{x^2} dx$

Put $x^2 = t \rightarrow 2x \, dx = dt$

$$\Rightarrow I = \frac{1}{2} \int t e^t \, dt$$

Integrating by parts, $\{u(t) = t, v(t) = e^t\}$

$$\Rightarrow I = \frac{1}{2} [t \int e^t \, dt - \int [(t)'] \int e^t \, dt] \, dt]$$

$$\Rightarrow I = \frac{1}{2} [te^t - \int [1 \cdot e^t] dt]$$

$$\Rightarrow I = \frac{1}{2} [te^t - e^t] + c$$

$$\Rightarrow I = \frac{1}{2} (x^2 - 1)e^{x^2} + c$$

$\therefore$ The correct answer is option (3).

Question 19

$$\int_0^{\frac{\pi}{2}} \frac{\cos x}{(1+\sin x)(2+\sin x)} dx =$$

Options:

A. $\log \frac{4}{3}$

B. $\log \frac{1}{3}$

C. $\log \frac{3}{4}$

D. None of these

Answer: A

Solution:

Calculation:

$$\text{Given, } I = \int_0^{\frac{\pi}{2}} \frac{\cos x}{(1+\sin x)(2+\sin x)} dx$$

$$\text{Put } \sin x = t \Rightarrow \cos x \, dx = dt$$

$$\text{and when } x = 0 \rightarrow t = 0$$

$$\text{when } x = \pi/2 \rightarrow t = 1$$

$$\Rightarrow I = \int_0^1 \frac{1}{(1+t)(2+t)} dt$$

$$\Rightarrow I = \int_0^1 \frac{1}{(1+t)} - \frac{1}{(2+t)} dt$$

$$\Rightarrow I = \log(1+t)|_0^1 - \log(2+t)|_0^1$$

$$\Rightarrow I = \log 2 - \log 1 - (\log 3 - \log 2)$$

$$\Rightarrow I = 2 \log 2 - \log 3$$

$$\Rightarrow I = \log \frac{4}{3}$$

$\therefore$ The correct answer is option (1).

Question 20

If p and q are non-zero constants, the equation $x^2 + px + q = 0$ has roots α and β , then the equation $qx^2 + px + 1 = 0$ has roots

Options:

A. α and $\frac{1}{\beta}$

B. $\frac{1}{\alpha}$ and β

C. $\frac{1}{\alpha}$ and $\frac{1}{\beta}$

D. None of these

Answer: C

Solution:

Given:

α and β are the roots of $x^2 + px + q = 0$, and

We have to find the roots of $qx^2 + px + 1 = 0$

Concept:

If α and β are the roots of quadratic equation $ax^2 + bx + c = 0$

Sum of root $(\alpha + \beta) = -b/a$

Product of root $(\alpha\beta) = c/a$

Calculation:

We have,

α and β are the roots of $qx^2 + px + q = 0$

$$\alpha + \beta = -p$$

$$\alpha\beta = q$$

Now for equation,

$$qx^2 + px + 1 = 0$$

Sum of root = $-p/q$

$$\Rightarrow \text{Sum of root} = (\alpha + \beta)/\alpha\beta$$

$$\Rightarrow \text{Sum of root} = 1/\alpha + 1/\beta \text{ ----(1)}$$

Product of root = $1/q$

$$\Rightarrow \text{Product of root} = 1/\alpha\beta$$

$$\Rightarrow \text{Product of root} = (1/\alpha)(1/\beta) \text{ ----(2)}$$

From equation (i) and (ii), we can say that

$1/\alpha$ & $1/\beta$ will be the roots of $qx^2 + px + 1 = 0$

$\therefore$ **Roots of equation $qx^2 + px + 1 = 0$ are $1/\alpha$ & $1/\beta$.**

Question 21

If $y = 1 - \cos\theta$, $x = 1 - \sin\theta$ then $\frac{dy}{dx}$ at $\theta = \frac{\pi}{4}$ is

Options:

A. - 1

B. 1

C. 1/2

D. $\frac{1}{\sqrt{2}}$

Answer: A

Solution:

CONCEPT:

- $\frac{d(\cos x)}{dx} = -\sin x$
- $\frac{d(\sin x)}{dx} = \cos x$

Differentiation of parametric functions:

If $x = f(t)$, $y = g(t)$, where t is a parameter, then $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)}$

CALCULATION:

Given: $y = 1 - \cos\theta$, $x = 1 - \sin\theta$

Here, we have to find $\frac{dy}{dx}$ at $\theta = \pi/4$

As we can see that, x and y are functions of θ

We also know that, if $x = f(t)$, $y = g(t)$, where t is a parameter, then $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}} = \frac{g'(t)}{f'(t)}$

Let's differentiate x and y w.r.t θ

$$\Rightarrow \frac{dy}{d\theta} = \frac{d(1 - \cos\theta)}{d\theta} = \sin\theta \text{-----(1)}$$

$$\Rightarrow \frac{dx}{d\theta} = \frac{d(1 - \sin\theta)}{d\theta} = -\cos\theta \text{-----(2)}$$

Now from (1) and (2) we can say that, $\frac{dy}{dx} = -\tan\theta$

$$\Rightarrow \left[\frac{dy}{dx} \right]_{(\theta=\frac{\pi}{4})} = -\tan\frac{\pi}{4} = -1$$

Hence, **option A** is the correct answer.

Question 22

If at any instant t , for a sphere, r denotes the radius, S denotes the surface area and V denotes the volume, then what is $\frac{dV}{dt}$ equal to?

Options:

A. $\frac{1}{2} S \frac{dr}{dt}$

B. $\frac{1}{2} r \frac{dS}{dt}$

C. $r \frac{dS}{dt}$

D. $\frac{1}{2} r^2 \frac{dS}{dt}$

Answer: B

Solution:

Concept:

Surface area $S = 4\pi r^2$

Volume of sphere $V = \frac{4}{3}\pi r^3$

Calculations:

Given, at any instant t , for a sphere, r denotes the radius, S denotes the surface area.

Surface area $S = 4\pi r^2$

Differentiating w.r.to t , we get

$$\Rightarrow \frac{ds}{dt} = 4\pi \cdot 2r \frac{dr}{dt}$$

$$\Rightarrow \frac{ds}{dt} = 8\pi r \frac{dr}{dt}$$

$$\Rightarrow \frac{dr}{dt} = \frac{1}{8\pi r} \frac{ds}{dt} \dots (1)$$

Volume of sphere $V = \frac{4}{3}\pi r^3$

Differentiating w.r.to t , we get

$$\Rightarrow \frac{dV}{dt} = \frac{4}{3}\pi \cdot 3r^2 \frac{dr}{dt}$$

$$\Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

From equation (1), we have

$$\Rightarrow \frac{dV}{dt} = 4\pi r^2 \frac{1}{8\pi r} \frac{ds}{dt}$$

$$\Rightarrow \frac{dV}{dt} = \frac{r}{2} \frac{ds}{dt}$$

Hence, if at any instant t , for a sphere, r denotes the radius, S denotes the surface area and V denotes the volume,
then $\frac{dV}{dt} = \frac{1}{2}r \frac{dS}{dt}$

Question 23

If a, b, c are non-zero real numbers, then the inverse of the matrix

$$A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix} \text{ is equal to}$$

Options:

$$\text{A. } \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$$

$$\text{B. } \frac{1}{abc} \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$$

$$\text{C. } \frac{1}{abc} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\text{D. } \frac{1}{abc} \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

Answer: A

Solution:

CONCEPT:

Inverse of a matrix: The Inverse of a $n \times n$ matrix is given by:

$$A^{-1} = \frac{\text{adj}(A)}{|A|} \text{ where } \text{adj}(A) \text{ is called adjoint matrix.}$$

Adjoint Matrix: If $B_{n \times n}$ is a cofactor matrix of matrix $A_{n \times n}$ then the adjoint matrix of $A_{n \times n}$ is denoted by $\text{adj}(A)$ and is defined as B^T . So, $\text{adj}(A) = B^T$.

CALCULATION:

$$\text{Given: } A = \begin{bmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{bmatrix}$$

$$\text{We know that, } A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$\Rightarrow |A| = [a(bc - 0) - 0(0) - 0(0)]$$

$$\Rightarrow |A| = abc$$

$$\Rightarrow \text{adj}(A) = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}^T$$

$$\Rightarrow \text{adj}(A) = \begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{\begin{bmatrix} bc & 0 & 0 \\ 0 & ac & 0 \\ 0 & 0 & ab \end{bmatrix}}{abc}$$

$$\Rightarrow A^{-1} = \begin{bmatrix} a^{-1} & 0 & 0 \\ 0 & b^{-1} & 0 \\ 0 & 0 & c^{-1} \end{bmatrix}$$

Question 24

The area bounded by curve $y^2 = x$, line $y = 4$ and y-axis is

Options:

A. $16/3$

B. $64/3$

C. $7\sqrt{2}$

D. None of these

Answer: B

Solution:

Concept:

The area bounded between the curve $x = f(y)$ and y-axis between $y = a$ and $y = b$ is

$$\int_a^b f(y) dy$$

Calculation:

Given curves, $y^2 = x$, line $y = 4$ and y-axis

$$\Rightarrow \text{Required area} = \int_0^4 y^2 dy = \left. \frac{y^3}{3} \right|_0^4$$

$$\Rightarrow \text{Required area} = 64/3$$

$\therefore$ The correct answer is option (2).

English

Question 25

Select the most appropriate synonym of the given word.

Eternal

Options:

A. Perpetual

B. Repetition

C. Prolonged

D. Look

Answer: A

Solution:

The correct answer is '**Perpetual**'.

Key Points

- The word '**Eternal**' means 'lasting forever or for a very long time (हमेशा के लिए या बहुत लंबे समय तक चलने वाला)'.
- The **synonyms** of the word '**Eternal**' are "abiding, ageless, perpetual, continuing, dateless, enduring, everlasting, immortal, imperishable, lasting, ongoing, perennial, perpetual, timeless, undying".
- **From the synonym** of the given word, we can say that the word '**Perpetual**' is the **most similar in meaning**.
- The word '**Perpetual**' means 'continuing forever in the same way (उसी तरह से हमेशा के लिए जारी)'.

Hence, the correct answer is **option 1**. Let's see the **meaning** of other given options:-

Words	Meaning
Repetition	the act of doing or saying something again (फिर से कुछ करने या कहने की क्रिया)
Prolonged	extend the duration of.
Look	to direct your eyes in order to see (देखने के लिए अपनी आंखों को निर्देशित करने के लिए)

Additional Information

- The **antonym** of the word '**Eternal**' is "transient".
- **Example of 'Eternal' in a sentence:**
 - Will you two never stop your *eternal* arguing!

Question 26

What is the antonym of the given word?

Gentle

Options:

- A. Mild
- B. Tender
- C. Harsh
- D. Merciful

Answer: C

Solution:

The correct answer is “**Harsh**”.

Key Points

- 'Gentle' denotes someone or something that is mild and kind, and does not cause harm or discomfort. (सौम्य)
 - **Example**-The deer in the forest were surprisingly **gentle**.
- 'Harsh' refers to something that is unkind, cruel, or more severe than is necessary. (कठोर)
 - **Example**-Her teacher's **harsh** words dampened her enthusiasm for learning.
- Hence, the option that is Opposite in meaning to the given word is 'Harsh'.

Therefore, the correct answer is '**Option 3**'.

Additional Information

- Let us have a look at the meaning of the remaining options.
 - 'Mild' implies a lack of harshness or severity. (सौम्य)
 - 'Tender' generally refers to something soft or delicate. In another context, it could also mean showing gentleness, kindness, and affection. (कोमल)
 - 'Merciful' denotes showing or exercising mercy. It refers to someone who forgives easily or is kind and forgiving. (दयालु)
-

Question 27

In the following question, out of the given four alternatives, select the alternative which best expresses the meaning of the Idiom/Phrase.

In black and white

Options:

- A. In writing
- B. In trouble
- C. In suspense
- D. In brief

Answer: A

Solution:

The correct answer is 'Option 1'.



Key Points

- "In black and white", the correct idiom meaning is "In writing" (कागज़ी रूप से).
- It refers to having something documented or written down.

Therefore, the correct answer is: 'In writing'



Additional Information

- In trouble(मुसीबत में) indicates a problematic situation, which doesn't match with the given idiom.
 - In suspense(सस्पेंस में) refers to a state of uncertainty or excitement, again, not connected to the idiom.
 - In brief(संक्षेप में) means something that is short and concise, not matching with the idiom.
-

Question 28

Find the word which is correctly spelt from the given options.

Options:

- A. Exemplary
- B. Indifferance
- C. Heirarchy
- D. Feudel

Answer: A

Solution:

The correct answer is, 'Exemplary'.



Key Points

- 'Exemplary' is the only correct spelling in the given options that means something/someone deserving of honour.
- Hence, the correct answer is 'option 1'

Therefore, the correctly spelt word is - 'Exemplary'.

Additional Information

- The correct spelling of the given words with their meaning.
 - **Indifference**- lack of enthusiasm.
 - **Hierarchy**- ranking, grading.
 - **Feudal**-relating to the social system of western Europe in the Middle Ages
-

Question 29

What is the verb form of the given noun?

Expansion

Options:

- A. Expandable
- B. Expanding
- C. Expand
- D. Expansionist

Answer: C

Solution:

The correct answer is "**Expand**".

Key Points

- '**Expansion**' is a noun denoting the action, process, or result of expanding.
- The **verb** form, which signifies the act of expanding something or becoming more extensive, is '**expand**.'
- Hence, the correct answer is '**Option 3**'.

Additional Information

- '**Expandable**' is an adjective resulting from the verb 'expand,' denoting the characteristic of being able to be made larger or wider. This word describes the possibility or ability of something to expand, not the action itself.
- '**Expanding**' is the present participle of the verb 'expand.' It represents an ongoing action of becoming larger or more extensive. Though it originates from the same verb, this is not the base verb form.
- '**Expansionist**' signifies an advocate of geographical or economic expansion. Even though it comes from the same root word, it is not the action form of the word 'expansion.'

Question 30

Convert the given sentence into Indirect speech.

Sunil said, "Where is Sunita?"

Options:

- A. Sunil asked where Sunita had been
- B. Sunil asked where Sunita is
- C. Sunil asked where Sunita was.
- D. Sunil said that where is Sunita.

Answer: C

Solution:

The correct answer is "**Sunil asked where Sunita was.**"



Key Points

- When turning a direct speech into an indirect one, we have to consider tense changes. The original question Sunil poses is in the present tense. However, because we're expressing this as indirect speech, it must be moved back a tense to become past tense - hence the "**is**" from the original sentence changes to "**was**".
- The punctuation used in the sentence is also crucial. In the direct speech, Sunil's statement is a question, indicated by the question mark. Once we change it to indirect speech, it becomes a statement about what Sunil asked. Consequently, the question mark should be replaced with a period in the indirect form.
- The form of the sentence changes as well in indirect speech. In direct speech, the question "where is Sunita?" is asked straightforwardly. However, in the indirect form, the sentence is more complex - it has to indicate that Sunil is the one who asked the question and what exactly the question was, all in a single statement. Therefore, the correct form should be "**Sunil asked where Sunita was.**"
- None of the other options align perfectly with these rules and considerations. Option A misgenders the tenses by using "had been" instead of "was," Option B retains the present tense "is," which is incorrect, and Option D fails to structure the indirect statement correctly, for it retains the "where is" order from the direct speech which is not appropriate for indirect speech.

Hence, the correct answer is '**Option 3**'.

Question 31

Transform the following sentence from active voice to passive voice:

Someone is following us

Options:

- A. We are being followed by someone
- B. Someone has been followed by us
- C. We were followed by someone.
- D. Someone are followed by us.

Answer: A

Solution:

The correct answer is **“We are being followed by someone.”**

Key Points

- The original sentence "Someone is following us" is in the active voice. The active voice in a sentence is the one where the subject performs the action. In this case, "Someone" performs the action of "following."
- In passive voice, the object of the active verb becomes the subject of the passive verb. Therefore, in transforming the sentence into passive, "us" becomes the subject, making "we" the proper word to start the sentence.
- The verb tense should remain the same when converting from active to passive voice. The active sentence uses present continuous tense ("**is following**"), so the passive form also needs to use present continuous tense. The passive form of present continuous is "**are being followed**" for "we", maintaining the same tense.
- The word "someone," which was the subject of the active sentence, now turns into the agent in the passive voice that follows the action "are being followed," retained by the use of the preposition "**by.**"
- Hence, Option 1, "We are being followed by someone," correctly transforms the active sentence into the passive voice without changing the meaning of the sentence.

Hence, the correct answer is **‘Option 1’**.

Question 32

Fill in the blank with correct option:

It _____ if you had not caught it.

Options:

- A. will break
- B. was breaking
- C. would have broken
- D. would break

Answer: C

Solution:

The correct answer is **would have broken**.



Key Points

- The given sentence is a third conditional type of question. In the third conditional sentence, the structure of the sentence is-
 - **If + Subject + had + V₃ + Object, Subject + Would + Have + V₃.**
- In the If-clause the structure used is "had not caught" this gives us a hint that this is the third type of conditional sentence.
 - Example-If you **had told** me about the meeting, I **would have come**.

So the correct sentence is- It **would have broken** if you **had not caught** it.

Question 33

In the given sentence, identify the segment which contains the grammatical error

Many a man have realized that real happiness lies in making sacrifices.

Options:

- A. Many a man
- B. have realized that

C. real happiness lies

D. in making sacrifices.

Answer: B

Solution:

The correct answer is '**Have realized that**'.

Key Points

- The given sentence is grammatically incorrect.
- Here, '**Has realized that**' should be used instead of '**Have realized that**'.
- With "**many a**" **singular verb** is used.
- Hence we should use '**has**' and not '**have**' when **asingular verb** is required in a sentence.
- *Example:*
 - *Many a man go to park every morning in my locality. (Incorrect)*
 - *Many a man goes to park every morning in my locality. (Correct)*
- Therefore, as per the points mentioned above, we find that the correct answer is **Option 2**.

Correct Sentence: *Many a man has realized that real happiness lies in making sacrifices.*

Additional Information

- **Each, every, everyone, somebody, someone, many a, more than one** etc. are **singular**.
 - Hence, they take singular verbs, singular nouns, and singular pronouns.
 - *Example:*
 - *Many a student have not done their class work. (Incorrect)*
 - *Many a student has not done his class work. (Correct)*
-

Question 34

Improve the underlined part. If no improvement is required, mark your answer as 'No improvement'.

When I get home, my children would be playing.

Options:

A. are playing

B. were playing

C. will have been playing

D. No improvement

Answer: C

Solution:

The correct answer is **“will have been playing”**.

Key Points

- The original sentence, **"When I get home, my children would be playing"** uses the incorrect tense. The phrase "would be playing" is the conditional continuous tense. It's used to describe actions that would happen under certain conditions. However, in the context of the sentence, the action is not conditional but rather a future prediction.
- Option 1 "are playing", makes the sentence present continuous - "When I get home, my children are playing". This is incorrect because it inaccurately suggests that the children are currently playing at the time of speaking, rather than predicting what they will be doing in the future when the speaker arrives home.
- Option 2 "were playing", makes the sentence past continuous - "When I get home, my children were playing". This is incorrect as it suggests the children were playing at the same time in the past when the speaker was getting home; it doesn't align with the future event of the speaker getting home.
- Option 3 **"will have been playing"**, makes the sentence future perfect continuous - "When I get home, my children will have been playing". This tense is used to describe an ongoing action that will be in progress up to some point in the future. Here, it suggests that the children will start playing before the speaker's arrival and continue to do so when the speaker arrives, which correctly matches the future context given in the sentence.

Hence, the correct answer is **‘Option 3’**.
